

UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR

Department of Electrical Engineering

LABORATORY REPORT

Subject:	EE-170L Computer Programming Lab
Lab No:	Lab 07
Lab Title:	Copying Arrays
Semester:	Semester 2
Session:	Spring 2026

Submitted By:

Name: Muhammad Adil

Registration No: BF25NWELE0669

1. OBJECTIVE

To understand and implement array copying in C++ by copying elements from one array to another element by element. To learn that arrays cannot be copied using a single assignment statement and must be copied individually using loops.

2. THEORY

An array is a collection of elements of the same data type stored in contiguous memory locations. In C++, there is no built-in statement to copy an entire array at once. Array elements must be copied individually, typically using a loop structure such as for or while.

When copying arrays, we need to: (1) Declare a source array and initialize it with values, (2) Declare a destination array of the same size, (3) Use a loop to copy each element from source to destination, (4) Verify the copied values by displaying them. The process involves accessing each element using its index and assigning the value from the source array to the corresponding position in the destination array.

3. LAB TASKS

3.1 Array Copying Program

Write a C++ program to declare an array of 4 integers, initialize it with values, copy it to another array, and display the copied array.

Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    // =====
    // Array Declaration Section
    // =====
    // Declare two arrays of size 4 to store integers
    int age[4];           // First array: stores original age values
    int same_age[4];      // Second array: will store copied values
    int i = 0;           // Loop counter variable, initialized to 0

    // =====
    // Array Initialization Section
    // =====
    // Assigning values to each element of the age array
    age[0] = 23;          // First element: age of person 1
    age[1] = 34;          // Second element: age of person 2
    age[2] = 64;          // Third element: age of person 3
    age[3] = 75;          // Fourth element: age of person 4

    // =====
```

```

// Array Copying Section
// =====
// Using for loop to copy elements from age to same_age
// Loop runs from i=0 to i<4 (covers indices 0,1,2,3)
for (; i < 4; i++)
{
    same_age[i] = age[i]; // Copy each element individually
                          // same_age[0]=age[0], same_age[1]=age[1],
etc.
}

// =====
// Output Display Section
// =====
// Display the copied array elements with tab separation
cout << "Original Array (age):    ";
for (i = 0; i < 4; i++)
{
    cout << age[i] << "\t";      // Display original array
}
cout << endl;

cout << "Copied Array (same_age): ";
for (i = 0; i < 4; i++)
{
    cout << same_age[i] << "\t"; // Display copied array
}
cout << endl;

return 0; // Return 0 to indicate successful execution
}

```

Expected Output:

```

Original Array (age):    23      34      64      75
Copied Array (same_age): 23      34      64      75

```

Explanation:

This program demonstrates that arrays in C++ cannot be copied using a single assignment statement like `same_age = age`. Instead, each element must be copied individually using a loop. The for loop iterates through each index from 0 to 3, copying the value from the source array (`age`) to the destination array (`same_age`) at the same index position.

4. CONCLUSION

This lab demonstrated that arrays in C++ cannot be copied using a single assignment statement. Instead, each element must be copied individually using a loop. The for loop iterates through each

index, copying the value from the source array to the destination array. This fundamental concept is essential for array manipulation in C++ programming. Understanding element-by-element copying is crucial for operations like array duplication, backup creation, and passing array data between different parts of a program.